

Estimating the Euro Area output gap through a Non-stationary DFM and a BVAR model

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- Total of 28 both macroeconomic and financial quarterly time series from the Euro area, from Q1 1999 to Q4 2019
- Accounted for the Foreign Block, through a block-exogenous matrix
 - 2 US variables have been included, to account for foreign influence
 - Restriction following the assumption of "small economy" for Europe
- No use of Covid data
- Preliminary transformation to induce raw data to be stationary
 - The particular transformations are model specific

2 steps:

- 1 isolate common dynamics in series through DFM;
- 2 PCA on factors to decompose them into long term trend and short term cycle (*trend-cycle decomposition*).

We'll get:

- *POTENTIAL OUTPUT* given by the deterministic and the stochastic long-run trend;
- *OUTPUT GAP* given by the difference between what is observed and the potential, i.e. due to the cycle.

Barigozzi-Luciani model specification

Step 1 - DFM

- assumptions on deterministic trend $\mathcal{D}_{it}$ for 2 groups of variables (n_b and $n - n_b$);
- $\mathcal{D}_{it} = b_{it-1} + \mathcal{D}_{it-1} + \epsilon_{it}$ & $b_{it} = b_{it-1} + \eta_{it}$;
- overall the model is:

$$y_{it} = \mathcal{D}_{it} + \lambda_i'(L)\mathbf{f}_t + \xi_{it} \quad (1)$$

Step 2 - PCA on $\mathbf{f}_t$

- $\mathbf{f}_t$ (q factors) are cointegrated of order d (s.t. $\beta'\mathbf{f}_t \sim I(0)$) and there are r common trends;
- assuming r common trends are linear combination of q factors (s.t. $\tau_t = \mathbf{B}'\mathbf{f}_t$), we get the d stationary vector $\mathbf{c}_t = \beta'\mathbf{f}_t$;
- overall $\mathbf{f}_t = \psi_1\tau_t + \psi_2\mathbf{c}_t$ where τ_t is trend & $\mathbf{c}_t$ is cycle;
- assuming τ_t & $\mathbf{c}_t$ contemporaneously not correlated $\implies \mathbf{B} = \beta_\perp$, s.t. $\beta_\perp'\beta = \mathbf{0}$.

Barigozzi-Luciani estimation

State-space form representation of the model: *observation equation* (1) and *transition equation* (2):

$$\mathbf{f}_t = \mathbf{A}(L)\mathbf{f}_{t-1} + \mathbf{u}_t \quad (2)$$

Initialization: get $\hat{\lambda}_i$ through PCA on differenced data; project y_{it} onto $\hat{\lambda}_i$ and get $\hat{\mathbf{f}}_t$; run (2) and get $\hat{\mathbf{A}}(L)$.

EM procedure:

- **E-step**: run Kalman filter starting with above estimates (from 1 to t); refine estimates through Kalman smoother (going from T to 1);
- **M-step**: build expected log-likelihood and maximize it wrt parameters of interest.

PCA on $\hat{\mathbf{f}}_t$:

- 1st PC is the *trend* (captures largest variation in factors);
- remaining components are the *cycle*.

Our VAR is estimated via Bayesian methods based on the Bayesian rule:

$$\pi(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{f(y)} \quad (3)$$

- The purpose of bayesian inference is to obtain $\pi(\theta|y)$.
- $\pi(\theta|y)$ reflects both prior beliefs and evidence from data.

Minnesota-type prior:

$$\mathbb{E}[\phi_i^{jk}] = 0 \quad (4)$$

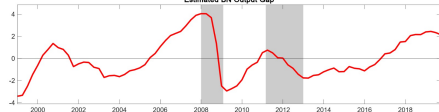
$$\text{Var}[\phi_i^{jk}] = \begin{cases} \frac{\lambda^2}{i^2}, j = k \\ \frac{\lambda^2}{i^2} \frac{\sigma_j^2}{\sigma_k^2}, \text{otherwise} \end{cases} \quad (5)$$

- the higher the lag i , the lower the shrinkage.
- σ_j^2, σ_k^2 are unknown and estimated
- λ regulates the level of shrinkage.
- ϕ_i^{jk} is the slope coefficient of the i^{th} lag of variable k in the j^{th} equation of the VAR.

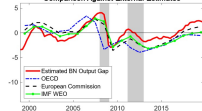
- **Normal independent Wishart prior** because of the Block-exogenous structure $\rightarrow$ the posterior cannot be analytically derived
- **MCMC Gibbs sampler method** for approximating the posterior distribution (**15000** draws are performed, **7000** are burned)
- **Block-exogenous structure on the VAR** for the foreign sector
- λ of 0.2 as baseline.

BL and MW estimates: a comparison with the benchmarks

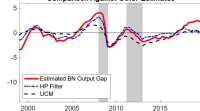
Estimated BN Output Gap



Comparison Against External Estimates



Comparison Against Other Estimates

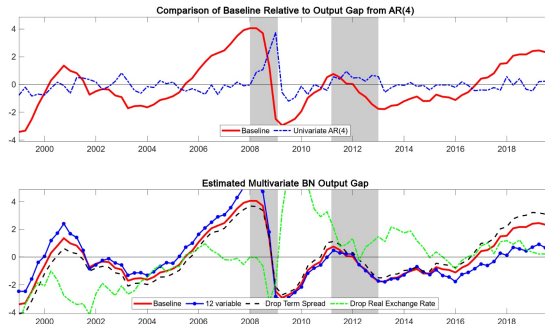


Output Gap - Level



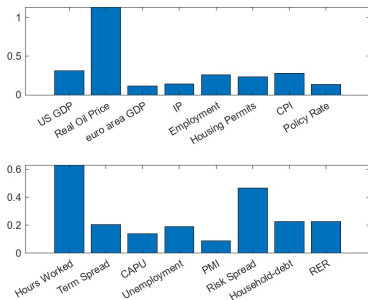
- MW estimates are qualitatively akin to other statistical methods, whilst they are less similar to other kind of methods.
- Generally similar qualitative behaviour (e.g when peaks and trough occur), significant quantitative differences.
- Computational efficiency is a stark difference between the models. This can severely limit the possibility of experimenting with different setups. (MW code takes at least 8h to run, BL takes few minutes).
- BL method is more suitable for data-rich environments. Morley and Wong performs better when only few series are available.

Drop Variables



- AR(4) is always around 0 with one significant spike after the 2008 crisis. This is evidence for the Household debt positive impact.
- There is a general qualitative similarity between the model, but some quantitative differences. This is absent in Wong, but robust to changing dropping household debt. It's imputable to dropping covid-data. This also applies to why we see difference when dropping real exchange rate.
- This strengthens our claim that financial variables are important in this empirical setting.

Informational content of variables



- We decompose the output gap into sources of forecast errors
- Almost all variables seems to contribute non negligibly to the output gap estimation
- Hours worked account for a large share of information content, different from Unemployment when using US data
- Few variables are less informative than they were in the original paper. This can be due both to the exclusion of Covid data or to the inclusion of Household debt